HACKASTONE 2026 · GRAND FINAL · AMSTERDAM

Grand Final Preparation Guide



Synapse by Team EduNova. The judging criteria, booth notes for every member, and a script and study guide for the pitchers.

Universiti Teknologi PETRONAS, Malaysia
Dutch Conference on AI for Education, VU Amsterdam, 29 October 2026

Read this first

Three parts: the judging criteria (what the panel scores), booth notes (for every member, all day), and a script and study guide for the pitchers (one block per slide).

1. **Part 1. The judging criteria**: what each criterion wants, what to show, proof we hold, and what to watch out for.
2. **Part 2. Booth notes**: 10-second and 60-second pitches, demo flow, key numbers, glossary, prepared answers, honest limits, checklist.
3. **Part 3. For the pitchers**: suggested script and an explanation for each of the 24 slides, with core and backup marked, plus a Q&A drill by criterion.

The event at a glance

Event	hackAstone 2026 Grand Final, inside the Dutch Conference on AI for Education, Vrije Universiteit Amsterdam
Date	29 October 2026 (conference 29 and 30 October; winners announced 30 October)
Our slot	10 minute presentation and 5 minute Q&A per the shortlist email. The competition site says 8 minutes and 2 minutes of Q&A. Plan for 8 and confirm with the secretariat
Booth	A booth for each team, staffed through the day, with a demonstration of up to 10 minutes including Q&A
Theme	Agentic AI for Education
Prizes	First S$1,000 (max 5), Second S$600 (max 10), Third S$350 (max 15), Merit S$200 (max 20), each with a certificate
Finalists	27 teams from about 20 countries, including our sister UTP team Lyceum
Dress	University uniform or team shirts, plus a banner or flag with the university identity
Team	EduNova, 8 UTP students: Por Jie Hao, Fakhrul Azim, Azrain Shawn, Khaidhir, Alif Fadhil, Adam Iman, Danish Iskandar, Nur Farah. Advisor: Assoc. Prof. Ts. Dr. Ahmad Sobri bin Hashim

Sources: the shortlist email, the competition site's Grand Final, Finalist and Criteria pages, and the team's grand-final deliverables plan.

Part 1. The judging criteria

The four criteria are equally weighted. The deck is built so a judge skimming sees all four. Source: the competition site's criteria page. The site lists no weights or sub-points beyond the one focus line per criterion.

1. Innovation and novelty Originality of concept

What the judges want. Original thought and a meaningful departure from existing solutions in the problem domain.

What to show.

- The one sharp idea: **the AI designs the experiment and never supplies a number.** Audited parts supply the science.
- The contrast with what judges have already seen: a chatbot with a plotting library, a one-shot "generate a UI" tool, a fixed simulation library.
- The trust mechanism as the novelty: only the verifier can mint a deliverable, so shipping an unverified lab is a code error, not a skipped check.

Proof we hold. Every lab is run in a real browser that moves every control and compares every number with an independent calculation. Planted-bug test: 219 of 224 refused, 0 of 30 false alarms. 20 of 20 hostile attacks blocked.

> **Watch out**
> Do not say "the first AI virtual lab". Claim the combination: verified, textbook-grounded, honestly labelled, generated on demand.

Carried by slides: 6, 9, 10, 11, 12, 15

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2. Technical readiness level Build quality and execution

What the judges want. A mature, robust, working implementation: engineering integrity, completeness of the build and demonstrable performance.

What to show.

- A **live or recorded run**, not slides of a system: one question typed, a lab built and checked, a student using it.
- The verification stage actually running (progress card, label, citation).
- Honest measured numbers with their limits printed beside them.

Proof we hold. 30 verified Environments (Physics 14, Chemistry 9, Biology 7). Kit of 34 audited building blocks and 17 science models. Real-browser checks, exact sandbox, one-file offline copy, phone-first 360 px layout, Present mode. A paid run measured about US$0.027 per new lab on 3 questions and 13 of 14 unseen questions built at about US$0.057 each.

> **Watch out**
> Nothing is deployed publicly yet: demo from the laptop and the offline file, not a public URL. Venue wifi may fail: offline lab first, recorded video second. Cost numbers are small samples, say so.

Carried by slides: 8, 11, 13, 14, 15, 17, 20

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3. Commercial viability Market potential

What the judges want. Potential to become a sustainable commercial venture or market-ready product, with a coherent value proposition and a credible pathway to adoption.

What to show.

- Who pays: students use it free; schools, districts, ministries and publishers are the paying route.
- Cost to serve: about 3 to 6 US cents to build a lab, about nothing to reopen one (the library is consulted first).
- Why a school adopts it: no apparatus needed, works offline on phones and school PCs, aligned to the national curriculum, every lab cites a textbook page.
- Why it compounds: each new audited model makes every later question in that topic cheaper and safer.

Proof we hold. Chapter packs and the offline file give a teacher a no-setup route. A teacher review lifts a lab from kit-verified to reviewed, which is the product a licence would sell.

> **Watch out**
> This is our thinnest criterion. We have **no pilot schools and no revenue**. Say "pilot is the next step". Do not invent revenue, schools or market-size figures. If the team has any real conversations or letters of interest, add them to slide 21.

Carried by slides: 12, 19, 21, 22

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4. Educational significance and applicability Pedagogical impact

What the judges want. Contribution to learning outcomes and capacity for meaningful deployment in real educational contexts.

What to show.

- The problem in a concrete image: a student who can recite osmosis and still believes water moves toward "more water". A quiz never catches it. A lab does.
- The pedagogy inside every lab: predict first, graded hints, a concept check, a worked solution, an exam-style question, and an honest trust label beside the textbook source.
- Fit to the context: Malaysia's reform toward reasoning, and the lab-access gap in rural and low-income schools.

Proof we hold. A textbook citation for each lab, a hint ladder, concept checks, a misconception probe, a practical card and lab report, an exam mirror. Progress stays on the student's device with no account.

> **Watch out**
> **No learner study has been run**, so claim no learning gains. **No teacher has reviewed the content.** The Bahasa Melayu strings are unreviewed drafts. Say all three before a judge finds them.

Carried by slides: 3, 4, 5, 7, 16, 18, 19, 22

Settle these first

Topic	What the sources say	Safe plan
Presentation length	The shortlist email says a 10 minute presentation plus 5 minutes of Q&A. The competition site's Grand Final page says "max 8 min" presentation, "2 mins QnA", and a demo of up to 10 minutes including Q&A at the booth.	Build the core story to fit 8 minutes (the "core" slides marked in Part 3). Keep the rest as backup. Ask the secretariat which is right. A news post dated 9 Oct announced a Tentative Program which we could not open.
Name	The organiser says the project title on every form is EduNova. The poster says "Synapse by Team EduNova".	Team is EduNova, product is Synapse. Say "Synapse by Team EduNova".

Topic	What the sources say	Safe plan
Team list	The poster lists 8 names. The prelim report listed 7. The old CodeFest and video credits (Team Dang Wangi) belong to a different event.	Use the poster's 8. Never show the Dang Wangi credits.
Theme fit	The theme is "Agentic AI for Education".	Say plainly which steps are AI (route, plan, wording judge) and which are deterministic (checks, assembler, verifier, gate). It plans, builds and checks its own work before the student sees it.
The judges	A steering committee of education and AI researchers (VU, Windesheim, UCD, Turku, PH Zurich, Kyoto, NTU and others) plus NVIDIA.	Lead with pedagogy and evidence. Keep engineering to one diagram.
The competition	27 finalists, many of them education tutors built on agents. Our sister team Lyceum from UTP is also in the final.	Our difference in one line: the others generate text or UI. We generate an experiment whose numbers cannot come from the AI and are checked in a real browser.

Claim hygiene: say this, never that

Safe to say	Do not say
"Every number comes from an audited science model, checked against an independent reference."	"Our AI is never wrong" or "100 percent accurate".
"219 of 224 planted bugs refused, 0 of 30 false alarms. 20 of 20 hostile cases blocked."	"It catches every bug."
"About US$0.027 per new lab, measured on three questions. About US$0.057 on a 14-question run. A saved lab reopens for almost nothing."	"It costs nothing to run."
"Citations are re-checked word for word against the textbook at delivery."	"Reviewed by teachers."
"Nothing unverified reaches the student."	"Teacher approved" or "canonical".
"Works offline on a phone as a single file."	"Used by N students."
"Bahasa Melayu switch built, translations are drafts awaiting review."	"Available in Bahasa Melayu."
"A study with learners is our next step."	"Improves exam scores."

Slide coverage check

Criterion	Slides that carry it
Innovation and novelty	6 (the idea), 9 (what it builds), 10 (related work), 11 (pipeline), 12 (lanes), 15 (verification)
Technical readiness	8 (demo), 11, 13, 14 (Kit), 15, 17 (grounding), 20 (measured)
Commercial viability	12 (lanes and cost), 19 (reach), 21 (adoption), 22 (limits and next)
Educational significance	3, 4, 5, 7, 16 (honest label), 18 (pedagogy), 19, 22

Part 2. Booth notes (for every member)

Use this section during the booth day. The booth runs all day, with at least two people on it at all times, rotating so nobody misses the keynotes or the award ceremony.

The 10-second version

> "Synapse turns a science question into a live, verified experiment instead of a paragraph, so students without a lab can still run one. Want to try it?"

The 60-second pitch

> "Synapse is an AI science tutor that answers a question with a working experiment instead of a paragraph. In Malaysia the exam reform expects students to reason from experiments, but many rural and low-income schools cannot provide labs. Type any high-school science question and Synapse builds a small lab you can load, measure and test, on a phone, with no account, even offline. The twist: the AI designs the experiment but never supplies a number. Audited building blocks supply the science, a real browser checks every control and every number, and every lab shows its textbook source and an honest trust label. Try it yourself: type a question."

End every pitch with the offer to try it on the laptop. Then collect a name and email from anyone interested: judges, other advisors and conference attendees are the reason a booth is worth staffing.

The story in five lines

- The reform expects students to reason from experiments, but many schools have no lab.
- A chatbot answers in text. An AI-written simulation can be confidently wrong.
- Synapse builds a real interactive lab for any question, on any phone, offline.
- The AI designs it and never supplies a number: audited parts and a real-browser check make it correct.
- Every lab shows its textbook page and an honest label, including what has not been reviewed.

Booth demo flow

- Ask the visitor for a topic (or offer Hooke's law, titration or the cannonball).
- Use a lab already stored in the library on the laptop, so it opens in seconds with no model call.
- Show the **predict card** first, then drag the masses (or the stopcock, or the launch speed) and watch the readouts.
- Open a **hint** and answer the **concept check**.
- Click the label and show the **provenance**: what was checked, the source page, and the limits.
- Show **Present**, **Share** and **Save offline**, then the EN | BM switch (say the Malay is a draft).

Fallbacks, in order

1. The offline lab file on the laptop. 2. The recorded demo video. Do not promise a public link: the API stays private until the deploy gate is met, so Share only works between machines once deployed.

Key numbers (always say the limit with the number)

Number	What it is	The limit to say
30	verified Environments: Physics 14, Chemistry 9, Biology 7	Hand-written plans. No teacher review.

Number	What it is	The limit to say
34 / 17	audited Kit parts / science models	As on the poster. The repo manifest has grown since.
219 / 224	planted bugs refused, 0 of 30 false alarms	Regression baseline written by the same team. Not an unbiased rate.
20 / 20	hostile attack shapes blocked by two layers	Known shapes, written by the defenders.
US$0.027	per new lab, measured live, about 21 seconds	Sample of 3 questions.
13 / 14	unseen Biology and Chemistry questions built, no example shown to the AI (about US$0.057 each)	Small sample, one author.
92.6% / 83.0%	local router accuracy on subject / topic	Agreement with an LLM teacher. English only.
26	national textbook PDFs ingested locally	No passages in the repository.

Source: the Synapse status ledger and evidence notes dated 28 Oct 2026, the poster, and our CodeFest deck. Re-run the eval scripts if anything changed.

The four labs in one line each

Lab	What the visitor does	What to point out
Hooke's law (Physics)	Predicts, then drags masses onto the hanger.	Extension and force over extension update live; every number is from an audited model.
Acid-base titration (Chemistry)	Pipettes, swirls, opens the stopcock, reads the burette at eye level.	The indicator changes at the end point; reading at eye level matters.
Yeast fermentation (Biology)	Builds the anaerobic setup, runs the timer, compares with the control tube.	The control tube is why the result means something.
Newton's cannonball (Physics)	Raises the launch speed.	The path goes from falling back, to orbit, to escape.

Glossary for international visitors

Term	Meaning
Synapse	Our product. An AI science tutor that answers a question with a live, verified, interactive experiment.
EduNova	Our team name and the project title on the organiser's forms.
UTP	Universiti Teknologi PETRONAS, our university, in Perak, Malaysia.
SPM	Sijil Pelajaran Malaysia. The national exam at the end of high school, taken at about 17.
KSSM	Kurikulum Standard Sekolah Menengah. Malaysia's national secondary-school curriculum.
KBAT	Kemahiran Berfikir Aras Tinggi. Higher-order thinking skills: the reform from recall to reasoning.
Form 4 and Form 5	The last two years of Malaysian high school, ages 16 and 17. We say "high school" to international guests.
B40	The bottom 40 percent of Malaysian households by income.
DELIMa	The Ministry of Education's national digital learning platform.
PISA	The OECD's international assessment of 15-year-olds. We cite the 2022 science scores.
Amali	Malay for practical work. Our "practical card" is the aim, variables and apparatus of an experiment.
Bahasa Melayu (BM)	The Malay language. Our BM switch is built; the translations are drafts awaiting review.
Environment	What a student receives: one interactive lab with its source and label. (The word "env" is only for settings files.)

Term	Meaning
Kit	Our library of audited building blocks (34 parts, 17 science models), versioned and hash-locked.
Lane	How a lab was made: Instantiate (stored), Compose (a plan over Kit parts) or Free (a labelled last resort).
Trust label	model-checked, kit-verified, reviewed, canonical. The last two need a person.
Verifier, mint, gate	The verifier runs the lab and alone can mint the deliverable. The gate re-checks it on delivery.
LLM seam	The one place every paid model call passes through, with a spend cap.
BM25	A standard text-retrieval method we use to find matching textbook passages.

Questions and prepared answers

Assign one primary answerer per topic before the day so there is no pause and no two people start at once. Each answer should take 30 to 45 seconds.

Likely question	Honest answer	Primary answerer
How is this different from a chatbot with a plotting library?	A chatbot gives text, and a plotting library draws whatever the model asks. In Synapse the AI only plans. Audited parts supply every number, and a real browser checks each control and number before the student sees the lab.	
What stops it generating a physically wrong simulation?	Numbers never come from the model. Each science model has a written precondition, a textbook citation, and an independent hand-written reference, and is tested against about 180 planted bugs. A lab that fails verification is dropped, not patched.	
Has a real student used it?	Not yet in a study. We have built and technically verified it. A classroom pilot with a pre and post study is our next step. Do not stretch this.	
What does one student cost to serve?	A new lab costs about 3 to 6 US cents (measured on small samples). A saved lab costs almost nothing to reopen, because the library is consulted before any model is called.	
Why Malaysia, and does it generalise?	The lab-access gap and the exam reform toward reasoning are specific to Malaysia, so we started there. The method (audited parts, a cited source, a verifier) does not depend on the syllabus, so another curriculum needs new audited models and a new textbook corpus, not a new pipeline.	
Who owns the content, and how does it track the syllabus?	The textbooks belong to the Ministry and the publisher: we ingest them locally, never store passages in the repository, and only cite the page. Every lab is bound to a textbook quotation that the gate re-proves, so a changed textbook makes the citation fail and the lab has to be rebuilt.	
How do you know a misconception was corrected?	We do not claim that yet. The app has a misconception probe and concept checks, and a pre and post study with learners is planned, not done.	
Is it really agentic?	It routes, plans, judges its own wording, composes and verifies, with a deterministic gate that no agent can bypass. Only the planning step is a creative AI step; the rest is deliberately deterministic so it can be trusted.	
Which models do you use?	Claude Sonnet 5.5 plans and Claude Haiku 5.5 routes, behind a seam that caps spend. A small local classifier does the first routing step.	
What about student privacy?	No accounts and no student data on the server. Progress stays on the student's device, with an export code.	
Does it work offline and on a cheap phone?	Yes: a lab can be saved as one hash-checked file that runs with no network, and the layout is built for a 360-pixel screen with large touch targets.	
What if the AI service is down or too expensive?	Every paid call goes through a spend cap and is refused before sending if it would exceed it. If a step fails, the ladder is a stored lab, then a composed one, then a written answer: never a blank screen.	

Likely question	Honest answer	Primary answerer
Why no teacher dashboard?	A deliberate decision to keep scope honest: Present mode projects one lab to a class, and a dashboard is future work.	
Is it free? Can I use it in my class?	Students would use it free with no account. It is not publicly hosted yet, so today we can share an offline lab file. Leave us your email and we will be in touch.	
Which subjects and levels does it cover?	High-school Physics, Chemistry and Biology for Malaysia's national curriculum: 30 verified labs from a census of 53 topics. Other subjects and levels are future work.	
Can I try it after the conference?	The service is private until our deploy gate is met, so there is no public link yet. We can share an offline lab file, and we will email you when it opens.	
Is the code open source?	We have not chosen a licence or a public host yet. Those are open decisions for the team, so do not promise either.	
Did you build this yourselves, or did AI?	We designed and built it as a team, with AI assistance, and we say so on our poster. The verification and trust design are ours.	
Does it replace a teacher or a real lab?	No. It is for students who have no lab, and a teacher can review and project it. A real lab is still better when it is available.	

Judges ask sharper versions of these. The full judge Q&A drill, with bridge-back lines, is at the end of Part 3.

Tech stack cheat sheet (for technical visitors)

Layer	What we use
Generative core	Claude Sonnet 5.5 plans, Claude Haiku 5.5 routes, typed plans via tool use, BM25 retrieval, a small local classifier
Component Kit	TypeScript, esbuild, 34 hash-locked parts, 17 science models
Backend	Python 3.13, FastAPI, Pydantic v2, SQLAlchemy and SQLite
Frontend	Next.js, React, Tailwind, server-sent events
Verification	Playwright and headless Chromium; one shared TypeScript to Python schema
Security	Sandboxed iframe (allow-scripts only), strict CSP, no runtime CDN, hash re-checked in the browser

Honest limits (say these before a judge finds them)

- No learner study yet, so no outcome claims.
- No teacher has reviewed a lab, so labels stop at kit-verified.
- 30 Environments, from a census of 53 curriculum topics.
- Bahasa Melayu strings are drafts awaiting review.
- Not publicly hosted: the API stays private until the deploy gate is met.
- No real-phone or screen-reader pass yet.
- Planted bugs refused are 219 of 224, not all.

Day-of checklist

- Offline lab file and recorded demo video on every laptop; the deck as a PDF.
- Laptops charged, chargers, a European plug adapter (type C or F), a mobile hotspot and a power strip.
- Roll-up banner, UTP flag or banner, and team shirts or uniforms.
- A sheet or form to capture names and emails.
- Booth roster written down: two people at all times.
- Agree with Lyceum: matching UTP shirts, a shared UTP banner, and non-overlapping rosters.

Time slot	Person 1	Person 2

Part 3. For the pitchers

One page block per slide: the slide, an explanation to study, a suggested spoken script, and what to watch out for. The same script is in the speaker notes of the deck. Read it aloud once, then say it in your own words.

How to deliver

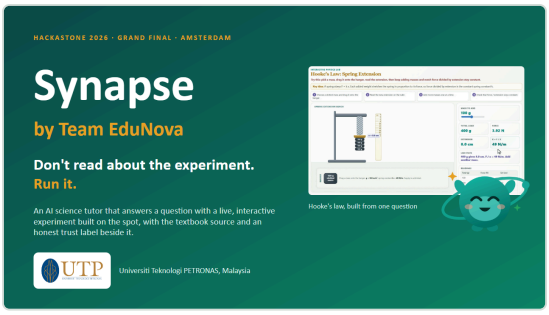
- One presenter for continuity, one person driving the demo. Fewer hand-offs read stronger.
- Rehearse against a clock. Rehearse to a minute or so under the limit: overrunning in front of a timer is avoidable.
- Say the limit with every number. Judges trust a team that states its own limits.
- Use "high school", not form levels, and explain Malaysian terms once (SPM, KBAT, B40) using the glossary in Part 2.
- Assume venue wifi fails: offline lab first, recorded video second.
- Wear the UTP shirt or uniform. Bring the banner or flag.

Short version: which slides to keep

If the slot is 8 minutes, present the **core** slides and keep the others as backup for Q&A: **1, 3, 4, 6, 7, 8, 11, 15, 16, 18, 20, 21, 22, 23, 24**. Everything else (2, 5, 9, 10, 12, 13, 14, 17, 19) stays in the deck as backup.

Slide 1 Cover

CORE



WHAT THIS SLIDE IS

The opening. It fixes three things in the audience's head: the product (Synapse), the team (EduNova, from Universiti Teknologi PETRONAS in Malaysia) and the one-line promise "Don't read about the experiment. Run it." The screenshot is the Hooke's law lab, the simplest lab to follow at a glance. Axi is the mascot and appears on every dark slide.

Criteria: All four

SUGGESTED SCRIPT

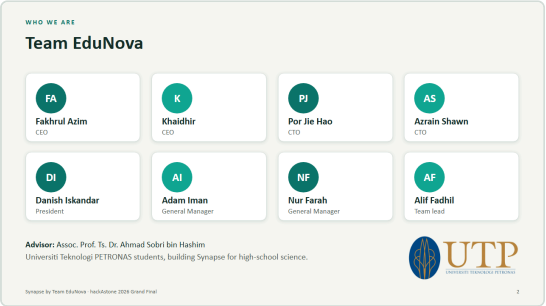
Good day, everyone. We are Team EduNova from Universiti Teknologi PETRONAS in Malaysia, and this is Synapse. Our line is: don't read about the experiment, run it. Synapse is an AI science tutor. You ask a science question and instead of a paragraph you get a live, interactive experiment, built on the spot, with the textbook source and an honest trust label beside it.

WATCH OUT

Name the presenter and the person driving the demo now. On official forms the project title is "EduNova"; on slides we say "Synapse by Team EduNova". Hold this slide about ten seconds, then move on.

Slide 2 Team

BACKUP



WHAT THIS SLIDE IS

Eight students and one advisor. Roles are as on the prelim deck. Alif Fadhil is listed as team lead from the organiser correspondence.

Criteria: Technical readiness (credibility)

SUGGESTED SCRIPT

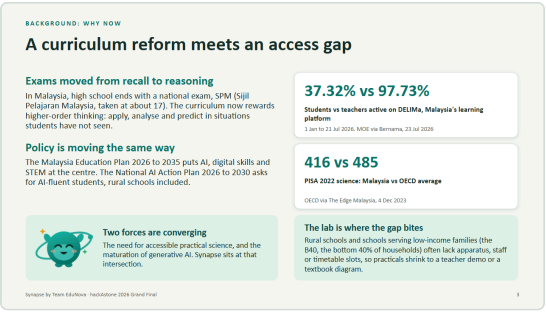
We are eight students from UTP, advised by Associate Professor Dr Ahmad Sobri bin Hashim. I will present, and my teammate will drive the demo.

WATCH OUT

Confirm every role and spelling with the owner of each name before the day. Skip this slide in a short version and just introduce the two speakers.

Slide 3 Background: a curriculum reform meets an access gap

CORE



WHAT THIS SLIDE IS

Sets the context for an international audience. Terms to know: KSSM is Malaysia's national secondary curriculum. KBAT (higher-order thinking skills) is the reform that moved teaching and exams from recall to reasoning. SPM (Sijil Pelajaran Malaysia) is the national exam at the end of high school, taken at about age 17. DELIMa is the Ministry of Education's national digital learning platform. PISA is the OECD's international assessment. B40 means the bottom 40 percent of households by income. The two stat cards show a gap between what policy wants and what students use. The bottom cards say why the lab is where it hurts.

Criteria: Educational significance

SUGGESTED SCRIPT

Some context. In Malaysia, students finish high school with a national exam called SPM, taken at about seventeen. The curriculum behind it has been reformed to test reasoning, not recall: apply, analyse, predict in situations you have not seen. Policy is pushing the same way: the national education plan and AI action plan both put AI and STEM at the centre, rural schools included. But look at the gap. Only 37 percent of students were active on the national learning platform, against 98 percent of teachers. Malaysia scored 416 in PISA science, against an OECD average of 485. And the lab is where this bites: rural schools and schools serving low-income families often lack apparatus, staff or timetable slots.

WATCH OUT

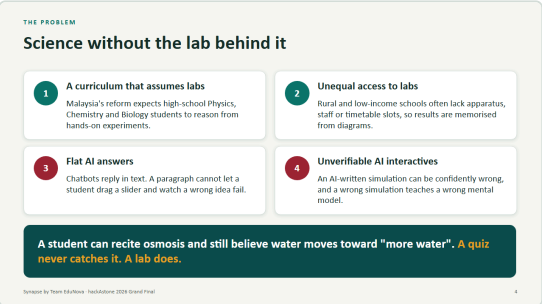
Source of the figures: our UTP CodeFest deck. DELIMa numbers are MOE via Bernama, 23 Jul 2026 (1 Jan to 21 Jul 2026). PISA 2022 is OECD via The Edge Malaysia, 4 Dec 2023. Re-check them if a judge asks. Do not say "Form 4 and 5": say "high school".

IF ASKED

Is this only a Malaysian problem? (The lab-access gap exists in many countries. We started where we know the curriculum and the gap.)

Slide 4 The problem: science without the lab behind it

CORE



WHAT THIS SLIDE IS

Four problems that stack. 1 and 2 are the education problem (the reform assumes labs, many schools cannot provide them). 3 and 4 are why existing AI does not fix it: a chatbot gives a flat paragraph, and an AI-written simulation can be confidently wrong, which teaches a wrong mental model. The dark banner is the emotional core of the whole pitch.

Criteria: Educational significance, Innovation

SUGGESTED SCRIPT

Four problems stack up. The reform assumes students work with real experiments. Many schools cannot provide them. A chatbot answers in text, and a paragraph cannot let you drag a slider and watch a wrong idea fail. And an AI-written simulation can be confidently wrong, which teaches a wrong mental model. Here is the one that matters most: a student can recite the definition of osmosis and still believe water moves toward "more water". A quiz never catches it. A lab does.

WATCH OUT

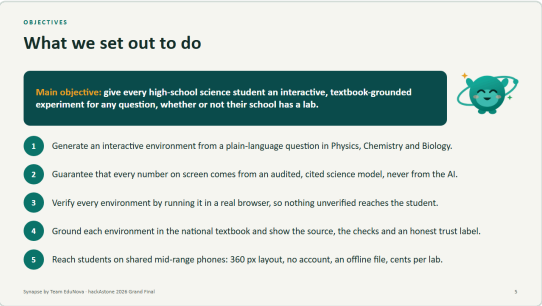
The osmosis example is a well-documented misconception in the science education literature. Do not claim Synapse has proved it fixes it: we have no learner study yet.

IF ASKED

How do you know the misconception example is real? (It is documented in the science-education literature. We do not claim to have fixed it.)

Slide 5 Objectives

BACKUP



WHAT THIS SLIDE IS

One main objective and five measurable goals. Each goal maps to something we built: plain-language generation, numbers only from audited models, real-browser verification, textbook grounding with an honest label, and phone-first delivery.

Criteria: Educational significance, Technical readiness

SUGGESTED SCRIPT

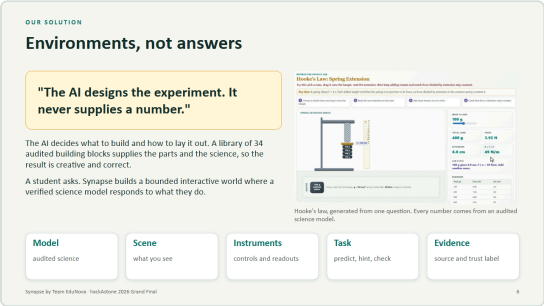
Our one objective: give every high-school science student an interactive, textbook-grounded experiment for any question, whether or not their school has a lab. Five goals follow. Generate the lab from plain language. Keep every number out of the AI's hands. Verify it in a real browser. Show the source, the checks and an honest label. And reach students on shared mid-range phones, with no account and about a few cents per lab.

WATCH OUT

A good slide to skip in a short version, because slides 6 and 15 say the same thing with more force.

Slide 6 Our solution: environments, not answers

CORE



WHAT THIS SLIDE IS

The central idea. The AI decides what to build and how to lay it out, but it never supplies a number. A library of audited building blocks supplies the parts and the science. Every environment has five parts: Model (audited science), Scene (what you see), Instruments (controls and readouts), Task (predict, hint, check) and Evidence (source and trust label). The quote card is the sentence to remember.

Criteria: Innovation and novelty

SUGGESTED SCRIPT

Here is the idea. The AI designs the experiment. It never supplies a number. The AI decides what to build and how to lay it out, and a library of thirty-four audited building blocks supplies the parts and the science, so the result is both creative and correct. Every environment has five parts: an audited science model, a scene you can see, instruments you control, a task such as predict, hint and check, and the evidence, meaning the source and an honest trust label.

WATCH OUT

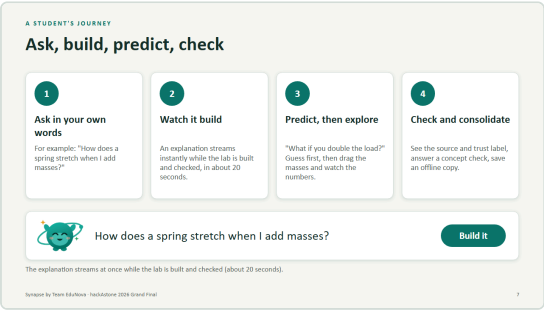
This is where a judge decides whether this is "a chatbot with a plotting library". The difference: the numbers cannot come from the model, and the page is checked in a real browser before the student sees it.

IF ASKED

Is that not just a chatbot with a plotting library? (No: the AI never supplies a number, and a real browser verifies the page.)

Slide 7 A student's journey

CORE



WHAT THIS SLIDE IS

The four-step flow the audience is about to see live: ask, watch it build, predict then explore, check and consolidate. The bar at the bottom is the actual entry point, a question box with a "Build it" button.

Criteria: Educational significance

SUGGESTED SCRIPT

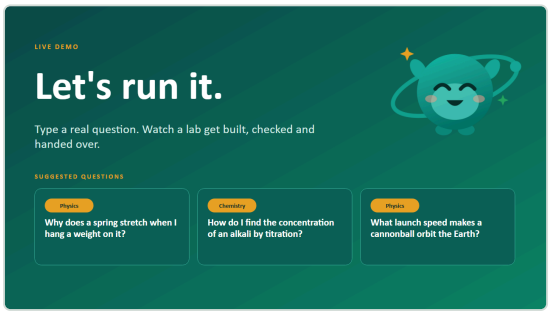
A student's journey is four steps. They ask in their own words, for example how a spring stretches when you add masses. They watch it build: an explanation streams at once while the lab is built and checked, about twenty seconds. Then they predict, because we make them guess before they see any numbers, and they explore by dragging the masses. Finally they check: they see the source and trust label, answer a concept check and can save an offline copy.

WATCH OUT

The "about 20 seconds" is for a newly composed lab. A lab already in our library opens in seconds with no model call.

Slide 8 Live demo

CORE



WHAT THIS SLIDE IS

The slide that hands over to the live demo. The three questions are suggestions: use labs already stored in the library on the demo laptop so no model call is needed.

Criteria: Technical readiness

SUGGESTED SCRIPT

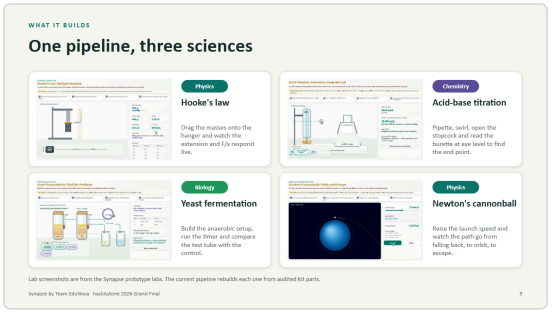
Let us run it. Type a real question. [Driver types] Watch what appears: the explanation streams, the progress card shows the lab being built and checked, and then the lab arrives with its label and its source.

WATCH OUT

Order: Hooke's law (predict, drag masses, hint, concept check), then titration, then the cannonball. Point at: the progress card, the "Kit verified" label, the textbook citation, Present, Share and Save offline. Fallbacks, in order: the offline lab file on the laptop, then the recorded demo video. Do not promise a public URL: the API stays private until the deploy gate is met.

Slide 9 What it builds: one pipeline, three sciences

BACKUP



WHAT THIS SLIDE IS

Four labs across Physics, Chemistry and Biology from the same pipeline: Hooke's law, an acid-base titration, yeast fermentation and Newton's cannonball. The screenshots are from the Synapse prototype labs shown on the poster; the current pipeline rebuilds each one from audited parts.

Criteria: Innovation, Technical readiness

SUGGESTED SCRIPT

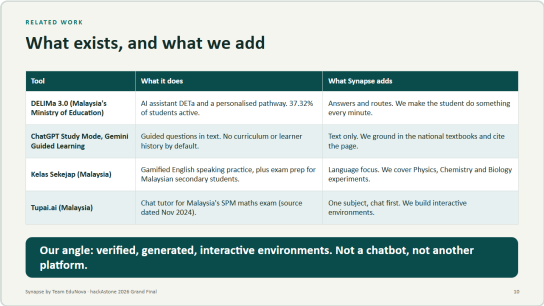
One pipeline, three sciences. Hooke's law: drag masses onto the hanger and watch the extension and force over extension respond. Titration: pipette, swirl, open the stopcock and read the burette at eye level. Fermentation: build the anaerobic setup, run the timer and compare with the control tube. And Newton's cannonball: raise the launch speed and watch the path go from falling back, to orbit, to escape.

WATCH OUT

Be honest if asked: these screenshots are from the prototype labs. The prototype titration used KOH and nitric acid; the current build uses sodium hydroxide and hydrochloric acid.

Slide 10 Related work: what exists, and what we add

BACKUP



WHAT THIS SLIDE IS

Four existing tools and the gap each leaves. DELIMa is Malaysia's national platform. Study Mode and Guided Learning are text-based. Kelas Sekejap and Tupai.ai are Malaysian apps (language practice; a maths chat tutor). The common gap: none makes the student run an experiment whose numbers are verified.

Criteria: Innovation and novelty

SUGGESTED SCRIPT

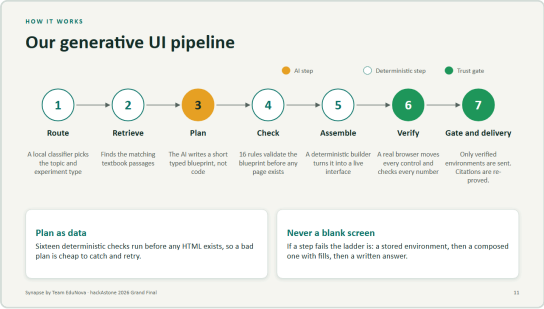
Other tools exist, and they are good. Malaysia's national platform answers questions and routes students. ChatGPT Study Mode and Gemini Guided Learning ask guided questions in text. A popular Malaysian app focuses on language, another is a maths chat tutor. What none of them does is make the student run an experiment whose numbers are verified. Our angle: verified, generated, interactive environments. Not a chatbot, not another platform.

WATCH OUT

Related-work facts come from our CodeFest deck. Re-check before quoting numbers. Do not criticise the Ministry's platform: say we complement it.

Slide 11 Our generative UI pipeline

CORE



WHAT THIS SLIDE IS

The seven steps. 1 Route: a small local classifier picks the topic and experiment type. 2 Retrieve: finds the matching textbook passages. 3 Plan (gold, the only AI step): the model writes a short typed blueprint, not code. 4 Check: sixteen deterministic rules validate the blueprint before any page exists. 5 Assemble: a deterministic builder turns the plan into a live interface. 6 Verify and 7 Gate and delivery (green, the trust gate): a real browser moves every control and checks every number, and only verified environments are sent, with citations re-proved. Colour code: gold is AI, outlined is deterministic, green is the trust gate.

Criteria: Innovation, Technical readiness

SUGGESTED SCRIPT

This is our generative UI pipeline, in seven steps. A small local classifier routes the question to a topic and experiment type. We retrieve the matching textbook passages. Then comes the only creative AI step: the model writes a short, typed blueprint. Not code. Sixteen rules check that blueprint before any page exists. A deterministic builder assembles the interface. A real browser then moves every control and checks every number. And the gate sends only verified environments, re-proving the citations on the way out. If any step fails, the student never sees a blank screen: we fall back from a stored lab, to a composed one, to a written answer.

WATCH OUT

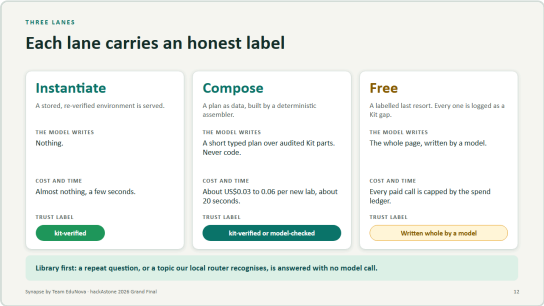
This answers "is it agentic?": it routes, plans and checks its own work, and a gate no agent can bypass. The poster says "16 rules"; the repository has since added more plan checks, so say "sixteen rules" only if you keep the poster wording.

IF ASKED

Which step is the AI? (Only step 3, Plan. The rest is deterministic.)

Slide 12 Three lanes, each honestly labelled

BACKUP



WHAT THIS SLIDE IS

Every lab is built in one of three lanes and carries that lane's label. Instantiate: a stored, re-verified lab (no model writes anything, almost free). Compose: the model writes a plan over audited parts as data (about 3 to 6 US cents, about 20 seconds). Free: a model writes the whole page as a labelled last resort, never canonical without a person and logged as a gap in our component Kit. The green banner is the cost story: a repeat question needs no model call.

Criteria: Innovation, Commercial viability

SUGGESTED SCRIPT

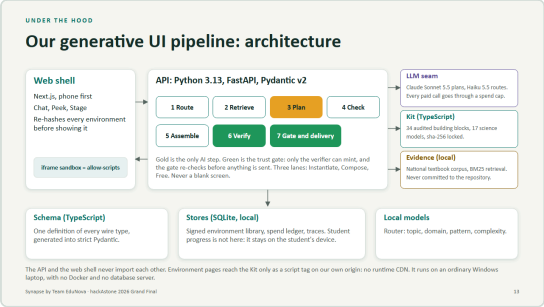
Every lab is built in one of three lanes and wears that lane's label. Instantiate serves a stored, re-verified lab: the model writes nothing, so it costs almost nothing. Compose is the main path: the model writes a plan over audited parts, and a deterministic assembler builds the page, for roughly three to six US cents. Free is a labelled last resort, where a model writes the whole page. It is never canonical without a human, and every one is logged as a gap in our component library. And because the library is consulted first, a repeat question needs no model call at all.

WATCH OUT

Cost figures: US$0.027 measured on 3 questions and US$0.057 per delivered lab on a 14-question run. Both are small samples, say so. The Free lane is built but we have not measured it.

Slide 13 Our generative UI pipeline: architecture

BACKUP



WHAT THIS SLIDE IS

The same seven steps, now as software. A phone-first web shell, a Python API that runs the pipeline, an LLM seam (Claude Sonnet 5.5 plans, Haiku 5.5 routes, spend capped), the TypeScript Kit of audited components, a local textbook corpus, and local stores. The API and the web shell never import each other; a generated page reaches the Kit only as a script tag on our own origin, with no runtime CDN.

Criteria: Technical readiness

SUGGESTED SCRIPT

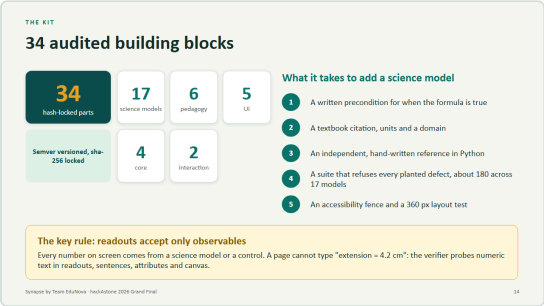
Here is the same pipeline as software. A phone-first web shell. A Python API running the seven steps. An LLM layer where Claude plans and a smaller model routes, with a spend cap on every paid call. A TypeScript library of audited parts. A local textbook corpus for evidence. And local stores, where student progress is deliberately not kept: it stays on the student's device. It runs end to end on an ordinary Windows laptop, with no Docker and no database server.

WATCH OUT

Technical judges may ask about the stack: Next.js, React and Tailwind on the web; Python 3.13, FastAPI, Pydantic v2 and SQLite on the API; Playwright with headless Chromium for verification; one shared TypeScript schema generated into Python. Do not read the stack aloud.

Slide 14 The Kit: audited building blocks

BACKUP



WHAT THIS SLIDE IS

The reason the science is right. Thirty-four audited, hash-locked parts: 17 science models, 6 pedagogy parts, 5 UI parts, 4 core and 2 interaction. Adding a science model needs a written precondition, a textbook citation with units and a domain, an independent hand-written reference in Python, a suite that refuses about 180 planted defects, and an accessibility fence. The key rule: readouts accept only observables, so a page cannot simply type "extension = 4.2 cm".

Criteria: Technical readiness, Innovation

SUGGESTED SCRIPT

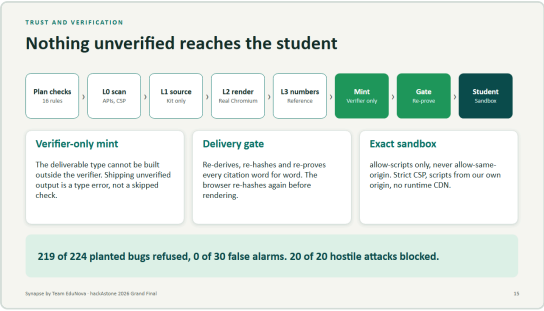
The Kit is why the science is right. Thirty-four audited, hash-locked parts: seventeen science models and the rest for pedagogy, interface and interaction. To add a science model we require a written precondition, a textbook citation with units, an independent hand-written reference, and a test suite that must refuse about a hundred and eighty planted bugs. And one rule matters most: readouts accept only observables. A page cannot just type "extension equals 4.2 centimetres". Every number comes from a model or a control.

WATCH OUT

The poster and our CodeFest deck say 34 parts and 17 models. The current build manifest lists more (62 components, 26 apparatus parts). If a judge asks for the repo count, say the poster counts the audited core Kit and the repo has grown since.

Slide 15 Nothing unverified reaches the student

CORE



WHAT THIS SLIDE IS

The trust mechanism, in order: plan checks, L0 scan (forbidden APIs, content security policy), L1 source (Kit parts only), L2 render (real Chromium moves every control across its range), L3 numbers (compared with an independent calculation), then the verifier alone can mint the deliverable, the gate re-checks it, and it reaches the student in a sandbox. Three guarantees: verifier-only mint (shipping unverified output is a type error, not a skipped check), a delivery gate (re-derives, re-hashes and re-proves citations), and an exact sandbox (allow-scripts only, never allow-same-origin).

Criteria: Technical readiness, Innovation

SUGGESTED SCRIPT

Nothing unverified reaches the student, and that is mechanism, not policy. After the plan checks, the page is scanned, its source must use only audited parts, and it is rendered in a real browser that moves every control across its range. Every number is compared with an independent calculation. Only then can the verifier mint the deliverable, and the type cannot be built anywhere else, so shipping unverified output is a type error, not a skipped check. The gate re-proves the citations on delivery, and the page runs in a locked sandbox. We tested this with planted bugs: 219 of 224 refused, with no false alarms, and 20 of 20 hostile attacks blocked.

WATCH OUT

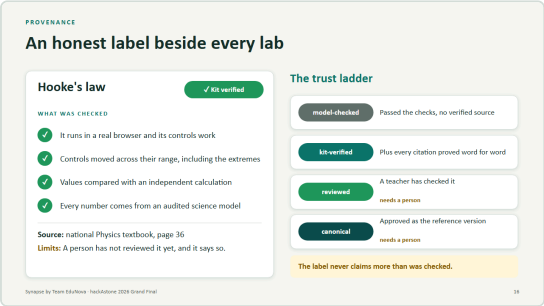
Say "219 of 224", never "every bug". These are our own regression baselines, written by the same team that wrote the verifier. The hostile attacks are known shapes written by the defenders.

IF ASKED

What if the verifier is wrong? (It is checked against independent references and planted bugs; 219 of 224 refused, and we report the misses.)

Slide 16 An honest label beside every lab

CORE



WHAT THIS SLIDE IS

Left: the provenance card a student sees, for Hooke's law: the label (Kit verified), what was checked, the textbook source and the limits. Right: the trust ladder. model-checked (passed the checks, no verified source) then kit-verified (plus every citation proved word for word) then reviewed and canonical, which need a person. The label never claims more than was checked.

Criteria: Educational significance, Innovation

SUGGESTED SCRIPT

Every lab shows its provenance: the label, what was checked, the source page and the limits. This is the card a student sees for Hooke's law. On the right is the trust ladder. Model-checked means it passed the checks but has no verified source. Kit-verified adds that every citation was proved word for word. Reviewed and canonical need a human. No teacher has reviewed our labs yet, so none is labelled reviewed, and the app says so. The label never claims more than was checked.

WATCH OUT

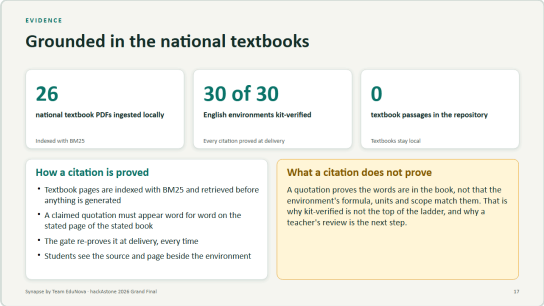
If a judge asks "what if the AI is wrong?", this slide is the answer. Be explicit that kit-verified is not a teacher's review.

IF ASKED

Has a teacher reviewed it? (No, so nothing is labelled reviewed, and the app says so.)

Slide 17 Grounded in the national textbooks

BACKUP



WHAT THIS SLIDE IS

We ingested 26 national textbook PDFs locally. All 30 English environments are kit-verified: each citation is a quotation that appears word for word on the stated page, re-proved by the gate at delivery. No textbook passage is stored in the repository. The cream card says what a citation does not prove: the words are in the book, not that the formula, units and scope match.

Criteria: Technical readiness, Educational significance

SUGGESTED SCRIPT

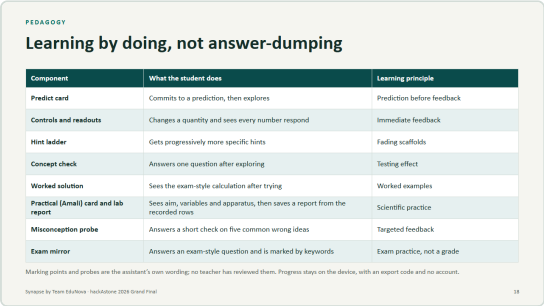
We ground before we generate. We ingested twenty-six national textbook PDFs, locally. Every one of our thirty English environments carries citations that the delivery gate re-proves: a quotation must appear word for word on the stated page. None of the textbook text is stored in our repository. And to be candid about the limit: a citation proves the words are in the book, not that our formula, units and scope match them. That is why kit-verified is not the top of the ladder, and why a teacher's review is our next step.

WATCH OUT

Copyright: textbooks stay on the local machine; the app shows a source and page, and a short quotation. If asked who owns the content, say the textbooks belong to the Ministry and the publisher, and we only cite them.

Slide 18 Learning by doing, not answer-dumping

CORE



WHAT THIS SLIDE IS

The pedagogy inside each lab, one row per component, with the learning principle behind it. Predict card (prediction before feedback), controls and readouts (immediate feedback), hint ladder (fading scaffolds), concept check (testing effect), worked solution (worked examples), practical card (Amali is Malay for practical work) and lab report, misconception probe (targeted feedback), exam mirror (exam practice, marked by keywords, not a grade).

Criteria: Educational significance

SUGGESTED SCRIPT

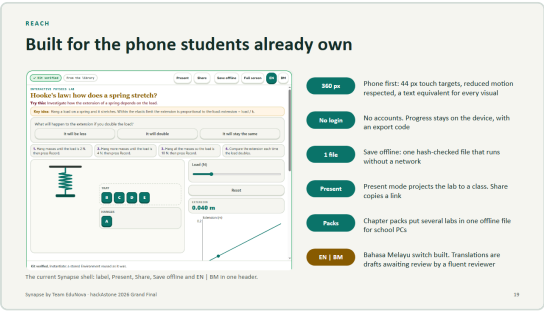
A lab only helps if students learn from it, so every environment carries pedagogy. The predict card makes the student commit before they see feedback. Hints fade from general to specific. A concept check uses the testing effect. A worked solution shows the exam-style calculation after they try. A practical card and lab report mirror real scientific practice, and an exam-style question gives practice, marked by keywords. Progress stays on the device, with no account.

WATCH OUT

Honest limit: the exam marking is keyword-based, not like a teacher, and the marking points and misconception probes are our own wording that no teacher has reviewed.

Slide 19 Built for the phone students already own

BACKUP



WHAT THIS SLIDE IS

Reach. The screenshot is the current web shell: the label, Present, Share, Save offline, Full screen and the EN | BM switch in one header. 360 px layout, 44 px touch targets, reduced motion, a text equivalent for every visual, no accounts, progress on the device, a one-file offline copy, Present mode to project a lab to a class, chapter packs for school PCs. Bahasa Melayu (the national language) switch is built; the translations are drafts awaiting review by a fluent reviewer.

Criteria: Educational significance, Technical readiness

SUGGESTED SCRIPT

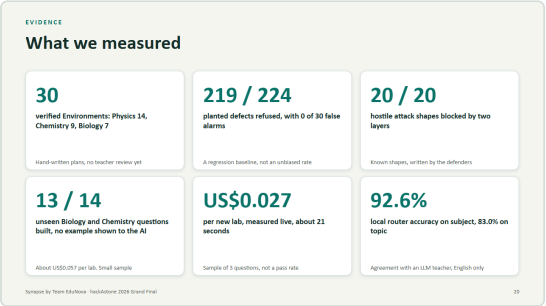
Built for the phone students already own. The layout is designed for a 360-pixel screen with large touch targets, reduced motion respected, and a text equivalent for every visual. No accounts: progress stays on the device. A lab can be shared as a link, projected to a class with Present mode, or saved as one offline file that runs with no network. Chapter packs put several labs in one file for school computers. We have also built a Bahasa Melayu switch, but those translations are drafts that still need a fluent reviewer.

WATCH OUT

Honest limits: no real-phone or screen-reader pass yet; keyboard and reduced-motion checks were done in a real browser. Say the Malay is draft, plainly.

Slide 20 What we measured

CORE



WHAT THIS SLIDE IS

Six numbers, each with its limit printed under it. 30 verified environments (Physics 14, Chemistry 9, Biology 7), hand-written plans, no teacher review. 219 of 224 planted defects refused, 0 of 30 false alarms (a regression baseline, not an unbiased rate). 20 of 20 hostile attack shapes blocked by two layers (known shapes). 13 of 14 unseen Biology and Chemistry questions built with no example shown to the AI, about US$0.057 per lab. US$0.027 per new lab, measured live, about 21 seconds, on 3 questions. Router accuracy 92.6 percent on subject and 83.0 percent on topic (agreement with an LLM teacher, English only).

Criteria: Technical readiness

SUGGESTED SCRIPT

What we measured, with the limits beside each number. Thirty verified environments across physics, chemistry and biology. 219 of 224 planted defects refused with no false alarms, which is a regression baseline we wrote ourselves, not an unbiased rate. 20 of 20 hostile attacks blocked. On cost: about two and a half US cents per new lab, measured live on three questions, and on a later fourteen-question run, thirteen produced a lab at about six cents each. Both are small samples. Our local router agrees with an LLM teacher 92.6 percent of the time on subject.

WATCH OUT

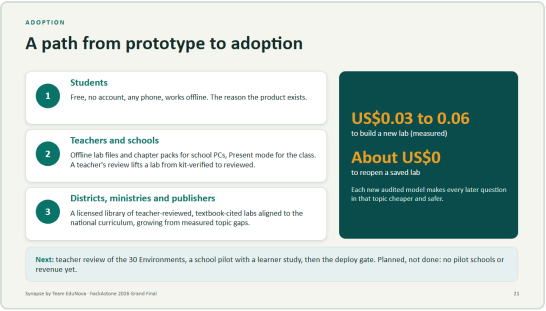
Source: the Synapse status ledger and evidence notes dated 28 Oct 2026. Re-run the eval scripts if anything changed. Never convert "13 of 14" into a pass rate.

IF ASKED

Is 13 of 14 a pass rate? (No: a small sample on unseen questions with no example shown to the AI.)

Slide 21 A path from prototype to adoption

CORE



WHAT THIS SLIDE IS

Our answer to "who pays". Students: free, no account. Teachers and schools: offline lab files, chapter packs and Present mode, and a teacher's review lifts a lab from kit-verified to reviewed. Districts, ministries and publishers: a licensed library of teacher-reviewed, textbook-cited labs aligned to the national curriculum, growing from measured topic gaps. Unit economics: about US$0.03 to 0.06 to build a lab, about nothing to reopen one. Each new audited model makes every later question in its topic cheaper and safer. This slide is a plan, not traction.

Criteria: Commercial viability

SUGGESTED SCRIPT

The logic is simple. Students pay nothing and need no account. Teachers and schools get offline lab files and chapter packs for school computers, Present mode for the class, and a teacher's review that lifts a lab from kit-verified to reviewed. The paying route is districts, ministries and publishers: a licensed library of teacher-reviewed, textbook-cited labs aligned to the national curriculum. It is cheap to run: about three to six US cents to build a lab, and about nothing to reopen one. And it compounds: each new audited model makes every later question in that topic cheaper and safer. This is a plan, not traction: we have no pilot schools or revenue yet.

WATCH OUT

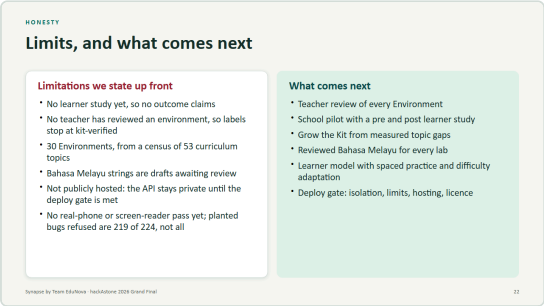
This is our thinnest criterion, so be straight. The content of this slide is our own planning, not sourced from a market study. Add any real conversations or letters of interest the team has. Do not quote market-size figures unless you can source them.

IF ASKED

Who pays? (Districts, ministries and publishers, for a teacher-reviewed library. No pilot or revenue yet.)

Slide 22 Limits, and what comes next

CORE



WHAT THIS SLIDE IS

We say our limits before a judge finds them. No learner study. No teacher review. 30 environments from a census of 53 curriculum topics. Bahasa Melayu strings are drafts. Not publicly hosted until the deploy gate is met. No real-phone or screen-reader pass; planted-bug refusal is 219 of 224, not all. Next: teacher review, a school pilot with a pre and post learner study, grow the Kit from measured gaps, reviewed Bahasa Melayu, spaced practice and difficulty adaptation, and the deploy gate.

Criteria: Educational significance, Technical readiness

SUGGESTED SCRIPT

We would rather tell you our limits than have you find them. We have no learner study yet, so we make no outcome claims. No teacher has reviewed a lab, so our labels stop at kit-verified. Thirty environments, from a census of fifty-three curriculum topics. The Malay is a draft. The API stays private until our deploy gate is met. And our planted-bug refusal is 219 of 224, not all. What comes next: teacher review of every lab, a school pilot with a pre and post learner study, growing the Kit from measured gaps, reviewed Bahasa Melayu, and the deploy gate.

WATCH OUT

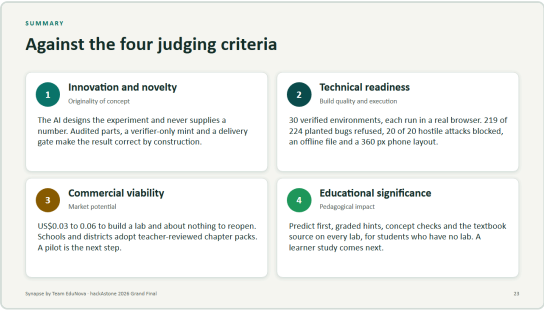
This is also the Q&A slide. If a judge asks "has a real student used it?" the honest answer is: not yet in a study, and that is our next step.

IF ASKED

Has a real student used it? (Not in a study. That is our next step.)

Slide 23 Against the four judging criteria

CORE



WHAT THIS SLIDE IS

A one-slide summary mapped to the four criteria: Innovation and novelty (the AI designs and never supplies a number, correct by construction), Technical readiness (30 verified environments, 219 of 224 planted bugs refused, 20 of 20 attacks blocked, offline file, phone layout), Commercial viability (cents to build, nothing to reopen, teacher-reviewed chapter packs, pilot next), Educational significance (predict first, graded hints, concept checks, source on every lab, a study next).

Criteria: All four

SUGGESTED SCRIPT

Against your four criteria. Innovation: the AI designs the experiment and never supplies a number, so the result is correct by construction. Technical readiness: thirty verified environments, each run in a real browser, with measured refusal rates. Commercial viability: cents to build, almost nothing to reopen, and a licensed, teacher-reviewed library as the route. Educational significance: predict first, graded hints, concept checks and the textbook source on every lab, for students who have no lab.

WATCH OUT

Keep this under 45 seconds. It is the summary the judges will score against.

Slide 24 Close

CORE



WHAT THIS SLIDE IS

The closing. The tagline repeats, the team names and UTP logo appear, and the call to action invites visitors to the booth.

Criteria: All four

SUGGESTED SCRIPT

Don't read about the experiment. Run it. Thank you. We are Team EduNova from Universiti Teknologi PETRONAS, and we will be at our booth all day, where you can type your own question and run a lab yourself. We are happy to take your questions.

WATCH OUT

Q&A: assign one primary answerer per topic before the day so there is no pause and no two people start at once. Prepared answers are in the booth notes section of this guide.

Q&A drill for the pitchers

The slot gives a few minutes of questions (5 per the email, 2 per the site). These are the questions judges most often ask in a final like this, grouped by criterion. Booth visitors ask casual versions: see the list in Part 2.

How to answer

1. Answer the question in your first sentence. **2.** State the honest limit. **3.** Bridge back to the strength in one line. Keep each answer to 30 to 45 seconds. One primary answerer per topic; the others stay quiet unless asked.

Innovation and novelty

Likely question	Suggested answer	Bridge back	Answerer
Isn't this just PhET with an LLM on top?	PhET is a fixed, hand-built library. Synapse builds the lab for the student's exact question, and the new part is the trust layer: the AI never supplies a number, a real browser verifies the page, and every lab carries a cited source and an honest label.	The novelty is the verified generation, not the simulation.	
Why not let the AI write the whole page?	Because a confidently wrong simulation teaches a wrong mental model. Our main path has the AI write a typed plan over audited parts. A whole-page lane exists as a labelled last resort, never canonical without a person, and every use is logged as a gap in our library.	Creative where that is safe, constrained where it matters.	
Is it really agentic?	It routes, plans, judges its own wording, composes and verifies, with a deterministic gate no agent can bypass. Only the planning step is a creative AI step; the rest is deterministic on purpose.	Agentic where it helps, deterministic where it must be trusted.	

Technical readiness

Likely question	Suggested answer	Bridge back	Answerer
How do you know the verifier itself is right?	It checks each number against an independent, hand-written reference, and we test it with planted bugs: 219 of 224 refused with no false alarms, and we report the five it missed. It is our own work and has not had an outside review yet, so we treat that rate as a regression baseline, not a guarantee.	We measure it rather than assume it.	
How long does a lab take, and does it scale?	A stored lab opens in seconds with no model call. A newly composed lab takes about 20 seconds and costs a few US cents. We have not load-tested at scale and the service is private for now, so we make no scale claims.	Library-first keeps cost and latency low as usage grows.	
What about prompt injection or unsafe output?	The model writes a plan as data, not code. A deterministic builder makes the page, the page is scanned, and it runs in a sandbox with no network or storage access. We tested 20 hostile attack shapes and blocked all 20; those are known shapes written by us.	The model never runs code on the student's device.	
What if a model provider is down or costs spike?	Every paid call passes through a spend cap and is refused before sending if it would exceed it. If a step fails the fallback is a stored lab, then a composed one, then a written answer. There is never a blank screen.	Failure is designed for, not hoped away.	

Commercial viability

Likely question	Suggested answer	Bridge back	Answerer
Who pays, and how do you earn if it is free for students?	Students are free. The paying route is districts, ministries and publishers licensing a teacher-reviewed, textbook-cited library aligned to the national curriculum. We have no pilot schools or revenue yet.	The next step is a school pilot with a learner study.	

Likely question	Suggested answer	Bridge back	Answerer
What is your moat? Could a big company copy this?	The hard part is not generation, it is the audited Kit, the verified library and the textbook grounding. Each new audited model makes every later question in that topic cheaper and safer, so the asset compounds and cannot be copied by prompting harder.	Trust and content are the moat, not the model.	
What would serving 500,000 students cost?	We have not measured it, so we will not quote a total. Because the library is consulted first, most requests would reopen a stored lab at almost no cost, and only a new question pays the few cents for a plan.	Do not invent a figure.	
Why would a ministry trust AI-made content?	It would not have to trust the model: every lab shows its textbook page and what was checked, and a teacher review is a defined step that lifts a lab from kit-verified to reviewed.	The label tells them exactly how far to trust each lab.	

Educational significance

Likely question	Suggested answer	Bridge back	Answerer
Is there evidence that predict-first and hints help?	The design follows well-established learning principles: prediction before feedback, fading scaffolds, the testing effect and worked examples. We have not measured learning gains with our own students yet. A pre and post study is planned.	We built on the evidence; we have not yet produced our own.	
Won't students just guess the prediction?	A guess still commits the student to a model, which makes the contradiction visible when the numbers respond. The concept check and probe then test the idea separately.	Being wrong first is the point.	
How do you know a misconception was corrected?	We do not claim that yet. The app has a misconception probe and concept checks. A study with learners is planned, not done.	Measure first, then claim.	
Is a phone lab as good as a real lab?	No, and we do not claim it is. It is for students who have no lab at all, and a teacher can use it before or after a real practical.	Something to do every minute beats a diagram.	
What about accessibility and weak connections?	The layout is built for a 360-pixel phone screen with large touch targets, reduced motion and a text equivalent for every visual, and a lab can be one offline file. We have done keyboard and reduced-motion checks in a real browser, but no real-phone or screen-reader pass yet.	State the checks done and the one not done.	
Why Malaysia, and does it generalise?	The lab-access gap and the exam reform are specific to Malaysia, so we started there. The method does not depend on the syllabus: a new curriculum needs new audited models and a textbook corpus, not a new pipeline.	Local problem, portable method.	

Team and next steps

Likely question	Suggested answer	Bridge back	Answerer
What happens in the next six months?	Teacher review of every lab, a school pilot with a pre and post learner study, reviewed Bahasa Melayu, growing the Kit from measured topic gaps, and the deploy gate: hosting, isolation, limits and a licence.	Review and pilot come before reach.	
What did you build yourselves?	The whole pipeline, the Kit, the verifier and the trust design, as a team of eight, with AI assistance, which we state on our poster.	Be direct and proud of the design choices.	